

PROVO, UT, USA

WMO: 725724

Lat: 40.219N

Lon: 111.723W

Elev: 1371

StdP: 85.91

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24174

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.9	-10.4	-16.4	1.1	-12.0	-13.9	1.3	-9.0	11.1	0.9	9.8	0.7	1.2	140	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.5	35.1	17.0	33.6	16.8	32.3	16.7	19.3	30.7	18.6	29.7	17.9	29.0	2.7	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.9	13.3	24.2	14.1	11.9	24.0	13.1	11.1	24.1	61.0	30.8	58.4	29.5	55.8	29.0	22.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
10.8	9.0	7.8	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-15.2	37.7	4.2	1.3	-18.2	38.6	-20.7	39.3	-23.0	40.0	-26.1	41.0
			-16.1	21.4	3.8	0.8	-18.8	22.0	-21.1	22.4	-23.2	22.9	-26.0	23.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	-1.2	1.2	6.2	9.6	14.6	19.8	24.6	23.3	17.4	10.3	3.8	-1.1
	DBStd	9.70	4.65	4.35	3.92	3.98	4.33	3.95	2.52	2.25	4.16	4.11	4.50	4.31
	HDD10.0	1372	347	248	129	54	10	1	0	0	3	47	191	343
	HDD18.3	3242	605	481	376	262	130	31	1	1	66	251	437	601
	CDD10.0	1654	0	1	12	44	152	296	453	413	225	55	4	0
	CDD18.3	482	0	0	0	1	14	76	196	156	38	1	0	0
	CDH23.3	5946	0	0	1	15	245	1041	2402	1736	473	32	0	0
	CDH26.7	2701	0	0	0	2	72	439	1248	787	149	4	0	0
(14) Wind	WSAvg	2.9	2.2	2.7	3.2	3.5	3.2	3.3	2.9	2.9	2.8	2.7	2.4	2.3
(15) Precipitation	PrecAvg	468	47	44	43	53	47	30	13	17	30	50	42	50
	PrecMax	579	97	85	84	116	139	95	34	37	63	116	78	134
	PrecMin	313	10	5	14	4	5	0	1	2	2	4	8	19
	PrecStd	75	29	21	20	26	32	22	9	10	18	27	20	26
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	14.8	22.2	25.5	32.2	35.7	37.7	35.5	32.5	27.1	18.2	12.4
		MCWB	5.7	7.2	10.4	12.4	16.5	15.6	17.9	16.9	16.0	12.8	8.6	6.2
	2%	DB	8.1	12.0	18.1	22.4	28.1	33.1	36.1	33.8	30.4	23.3	16.0	8.9
		MCWB	4.0	5.7	8.7	10.8	14.0	15.5	17.5	16.8	15.5	12.1	7.9	4.5
	5%	DB	6.8	9.6	16.1	20.0	26.2	31.7	34.2	32.5	28.2	21.1	13.0	7.2
		MCWB	3.3	4.7	7.7	10.2	13.5	15.5	17.2	16.9	15.3	11.6	6.5	3.3
	10%	DB	4.9	7.5	13.3	17.6	23.5	29.0	32.8	31.2	26.3	17.9	11.2	5.2
		MCWB	2.1	3.5	6.6	9.1	12.8	15.0	17.2	16.8	14.8	10.0	5.9	2.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.4	7.8	10.8	13.0	17.4	18.3	20.9	19.8	18.8	14.8	9.9	6.7
		MCDB	9.7	13.6	20.1	23.6	29.8	29.4	31.6	30.4	28.3	23.1	16.6	11.8
	2%	WB	4.7	6.2	9.5	11.5	15.4	17.1	19.7	18.8	17.4	13.0	8.4	4.9
		MCDB	7.9	11.0	17.5	20.7	26.0	28.9	31.8	30.0	26.9	21.3	14.6	8.4
	5%	WB	3.6	4.9	8.3	10.4	14.1	16.3	19.0	18.1	16.2	11.8	7.2	3.6
		MCDB	6.3	8.9	15.1	18.7	23.7	28.7	30.5	29.4	26.0	19.6	12.6	6.6
	10%	WB	2.4	3.8	6.9	9.5	13.1	15.6	18.3	17.5	15.2	10.5	6.2	2.4
		MCDB	4.9	7.2	13.0	17.1	22.8	27.7	29.5	28.7	24.8	17.7	10.6	5.1
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	9.0	11.2	11.9	13.6	15.5	16.5	15.8	14.9	13.2	10.3	8.1
		MCDBR	9.5	11.4	15.4	16.3	17.6	19.0	18.4	17.8	17.6	17.5	14.0	9.7
	5% WB	MCWBR	6.3	6.7	8.2	8.1	7.9	8.0	7.5	7.4	8.1	8.8	7.9	6.3
		MCWBR	8.9	10.5	14.5	15.4	16.1	17.3	16.2	15.6	15.9	16.0	13.1	9.2
(40) Clear-Sky Solar Irradiance	taub		0.251	0.264	0.280	0.309	0.317	0.313	0.347	0.343	0.306	0.284	0.262	0.250
		taud	2.473	2.506	2.528	2.455	2.493	2.537	2.473	2.480	2.564	2.598	2.587	2.507
	Ebn at Noon		935	969	983	966	959	958	922	920	941	934	920	910
		Edh at Noon	75	85	93	108	107	102	108	104	90	77	67	67
(44) All-Sky Solar Radiation	RadAvg		2.23	3.06	4.48	5.60	6.57	7.80	7.35	6.40	5.38	3.77	2.49	1.86
	RadStd		0.19	0.30	0.32	0.36	0.55	0.60	0.36	0.34	0.39	0.31	0.25	0.21

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page